1)
```
#include<stdio.h>
#include<stdlib.h>
int main(){
    int i,n,large;
    scanf("%d",&n);
    int *ptr=(int*)malloc(n*sizeof(int));
    for(i=0;i<n;i++)
    scanf("%d",&ptr[i]);
    large = ptr[0];
    for(i=0;i<n;i++){
        if(ptr[i]>large)
        large = ptr[i];

    }
    for(i=0;i<n;i++)
    printf("%d ",ptr[i]);
    printf("\nLargest element:%d",large);

}
```

Output:

```
5
10 56 34 78 23
10 56 34 78 23
Largest element:78
```

2)
```
#include<stdio.h>
int main(){
    int i,n,*ptr;
    scanf("%d",&n);
    int a[n];
    ptr = a;
    for(i=0;i<n;i++)
    scanf("%d",&*(ptr+i));
    for(i=0;i<n;i++)
    printf("%d ",*(ptr+i));
    printf("\nReversed array:\n");
    for(i=n-1;i>=0;i--)
    printf("%d ",*(ptr+i));
}
```

Output:

```
5
```

10 20 30 40 50
10 20 30 40 50
Reversed array:
50 40 30 20 10

3)#include <stdio.h>
#include <string.h>

```
struct Stud
{
   int rollno;
   char name[30];
   char gender;
};

int main()
{
   int n, i, j;
   struct Stud temp;

   scanf("%d", &n);

   struct Stud s[n];

   for(i = 0; i < n; i++)
   {
      scanf("%d", &s[i].rollno);
      scanf("%s", s[i].name);
      scanf(" %c", &s[i].gender);
   }

   printf("Student details:\n");

   for(i = 0; i < n; i++)
   {
      printf("%d %s %c\n", s[i].rollno, s[i].name, s[i].gender);
   }

   printf("\nSorting by name:\n");

   for(i = 0; i < n - 1; i++)
   {
      for(j = i + 1; j < n; j++)
```

```
        {
            if(strcmp(s[i].name, s[j].name) > 0)
            {
                temp = s[i];
                s[i] = s[j];
                s[j] = temp;
            }
        }
    }

    for(i = 0; i < n; i++)
    {
        printf("%d %s %c\n", s[i].rollno, s[i].name, s[i].gender);
    }

    return 0;
}
```

Output:

```
3
101 reena F
104 stacy F
105 bob M
Student details:
101 reena F
104 stacy F
105 bob M

Sorting by name:
105 bob M
101 reena F
104 stacy F
```

4)

```
#include <stdio.h>

struct Distance
{
    int feet;
    int inches;
};

int main()
{
```

```
    int n, i;

    scanf("%d", &n);

    struct Distance d[n];
    struct Distance sum = {0, 0};

    for(i = 0; i < n; i++)
    {
        scanf("%d %d", &d[i].feet, &d[i].inches);
    }

    for(i = 0; i < n; i++)
    {
        sum.feet += d[i].feet;
        sum.inches += d[i].inches;
    }

    sum.feet += sum.inches / 12;
    sum.inches = sum.inches % 12;

    printf("Total distance = %d feet %d inches",
           sum.feet, sum.inches);

    return 0;
}
```

Output:

```
3
5 7
3 12
4 8
Total distance = 14 feet 3 inches
```

5)

```
#include <stdio.h>

struct Complex
{
    int real;
    int img;
};
```

```
struct Complex add(struct Complex c[], int n)
{
   struct Complex sum = {0, 0};

   for(int i = 0; i < n; i++)
   {
      sum.real += c[i].real;
      sum.img += c[i].img;
   }

   return sum;
}

int main()
{
   int n;Prod

   scanf("%d", &n);

   struct Complex c[n], sum;

   for(int i = 0; i < n; i++)
   {
      scanf("%d %d", &c[i].real, &c[i].img);
   }

   sum = add(c, n);

   printf("Sum = %d + %di", sum.real, sum.img);

   return 0;
}
```

Output:

```
2
3 7
2 8
Sum = 5 + 15i
```

6)

```
#include <stdio.h>

struct Prod
{
```

```
    int id;
    char name[30];
    int qty;
    float price;
};

int addprod(struct Prod p[], int n)
{
    scanf("%d", &p[n].id);
    scanf("%s", p[n].name);
    scanf("%d", &p[n].qty);
    scanf("%f", &p[n].price);

    return n + 1;
}

void display(struct Prod p[], int n)
{
    int i;

    for(i = 0; i < n; i++)
    {
        printf("%d ", p[i].id);
        printf("%s ", p[i].name);
        printf("%d ", p[i].qty);
        printf("%.2f\n", p[i].price);
    }
}

int search(struct Prod p[], int n, int val)
{
    int i;

    for(i = 0; i < n; i++)
    {
        if(p[i].id == val)
        {
            return i;
        }
    }

    return -1;
}
```

```
int deleteprod(struct Prod p[], int n, int val)
{
    int i, pos;

    pos = search(p, n, val);

    if(pos == -1)
    {
        printf("Product not found\n");
        return n;
    }


    for(i = pos; i < n - 1; i++)
    {
        p[i] = p[i + 1];
    }

    return n - 1;
}

int main()
{
    struct Prod p[100];
    int n, choice, val;

    scanf("%d", &n);


    for(int i = 0; i < n; i++)
    {
        scanf("%d", &p[i].id);
        scanf("%s", p[i].name);
        scanf("%d", &p[i].qty);
        scanf("%f", &p[i].price);
    }

    while(1)
    {
        printf("\n1. Add Product\n");
        printf("2. Display Products\n");
        printf("3. Search Product\n");
```

```
        printf("4. Delete Product\n");
        printf("5. Exit\n");

        scanf("%d", &choice);

        switch(choice)
        {
            case 1:
                n = addprod(p, n);
                break;

            case 2:
                display(p, n);
                break;

            case 3:
                scanf("%d", &val);

                if(search(p, n, val) == -1)
                    printf("Product not found\n");
                else
                    printf("Product found\n");

                break;

            case 4:
                scanf("%d", &val);
                n = deleteprod(p, n, val);
                break;

            case 5:
                return 0;

            default:
                printf("Invalid choice\n");
        }
    }
}
```

Output:

```
2
101 phone 1 30000
102 earbud 2 1500
```

1. Add Product
2. Display Products
3. Search Product
4. Delete Product
5. Exit
1
103 tablet 1 50000

1. Add Product
2. Display Products
3. Search Product
4. Delete Product
5. Exit
2
101 phone 1 30000.00
102 earbud 2 1500.00
103 tablet 1 50000.00

1. Add Product
2. Display Products
3. Search Product
4. Delete Product
5. Exit
3
102
Product found

1. Add Product
2. Display Products
3. Search Product
4. Delete Product
5. Exit
4
101

1. Add Product
2. Display Products
3. Search Product
4. Delete Product
5. Exit
2
102 earbud 2 1500.00

103 tablet 1 50000.00

1. Add Product
2. Display Products
3. Search Product
4. Delete Product
5. Exit
5
7)#include <stdio.h>

```
int main()
{
    int n, temp, rem;
    int binary[32], i = 0;

    printf("Enter decimal number: ");
    scanf("%d", &n);


    temp = n;
    while(temp > 0)
    {
        binary[i] = temp % 2;
        temp = temp / 2;
        i++;
    }

    printf("Binary = ");
    for(i = i - 1; i >= 0; i--)
    {
        printf("%d", binary[i]);
    }


    temp = n;
    int octal[32], j = 0;

    while(temp > 0)
    {
        octal[j] = temp % 8;
        temp = temp / 8;
        j++;
    }
```

```
    printf("\nOctal = ");
    for(j = j - 1; j >= 0; j--)
    {
        printf("%d", octal[j]);
    }


    temp = n;
    char hex[32];
    int k = 0;

    while(temp > 0)
    {
        rem = temp % 16;

        if(rem < 10)
            hex[k] = rem + '0';
        else
            hex[k] = rem - 10 + 'A';

        temp = temp / 16;
        k++;
    }

    printf("\nHexadecimal = ");
    for(k = k - 1; k >= 0; k--)
    {
        printf("%c", hex[k]);
    }

    return 0;
}
```

Output:
Enter decimal number: 12
Binary = 1100
Octal = 14
Hexadecimal = C

8)#include <stdio.h>

```
struct Time
{
   int hours;
   int minutes;
   int seconds;
};

int main()
{
   struct Time t1, t2, diff;

   printf("Enter first time (hours minutes seconds): ");
   scanf("%d %d %d", &t1.hours, &t1.minutes, &t1.seconds);

   printf("Enter second time (hours minutes seconds): ");
   scanf("%d %d %d", &t2.hours, &t2.minutes, &t2.seconds);


   if(t1.seconds < t2.seconds)
   {
      t1.seconds += 60;
      t1.minutes--;
   }


   if(t1.minutes < t2.minutes)
   {
      t1.minutes += 60;
      t1.hours--;
   }

   diff.seconds = t1.seconds - t2.seconds;
   diff.minutes = t1.minutes - t2.minutes;
   diff.hours = t1.hours - t2.hours;

   printf("Difference = %d hours %d minutes %d seconds",
         diff.hours, diff.minutes, diff.seconds);

   return 0;
}
```

Output:

```
Enter first time (hours minutes seconds): 10 20 30
```

```
Enter second time (hours minutes seconds): 5 15 40
Difference = 5 hours 4 minutes 50 seconds
```

9)

```
#include <stdio.h>
#include <string.h>

struct Employee
{
    int id;
    char name[30];
    char department[30];
    float salary;
};

int addEmployee(struct Employee e[], int n)
{
    scanf("%d", &e[n].id);
    scanf("%s", e[n].name);
    scanf("%s", e[n].department);
    scanf("%f", &e[n].salary);

    return n + 1;
}

void display(struct Employee e[], int n)
{
    int i;

    for(i = 0; i < n; i++)
    {
        printf("%d %s %s %.2f\n",
               e[i].id,
               e[i].name,
               e[i].department,
               e[i].salary);
    }
}

int search(struct Employee e[], int n, int val)
{
    int i;
```

```
    for(i = 0; i < n; i++)
    {
        if(e[i].id == val)
        {
            return i;
        }
    }

    return -1;
}

int deleteEmployee(struct Employee e[], int n, int val)
{
    int i, pos;

    pos = search(e, n, val);

    if(pos == -1)
    {
        printf("Employee not found\n");
        return n;
    }

    for(i = pos; i < n - 1; i++)
    {
        e[i] = e[i + 1];
    }

    return n - 1;
}

int main()
{
    struct Employee e[100];
    int n, choice, val, pos;

    scanf("%d", &n);

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &e[i].id);
        scanf("%s", e[i].name);
        scanf("%s", e[i].department);
```

```
    scanf("%f", &e[i].salary);
}

while(1)
{
    printf("\n1. Add Employee\n");
    printf("2. Display Employees\n");
    printf("3. Search Employee\n");
    printf("4. Delete Employee\n");
    printf("5. Exit\n");

    scanf("%d", &choice);

    switch(choice)
    {
        case 1:
            n = addEmployee(e, n);
            break;

        case 2:
            display(e, n);
            break;

        case 3:
            scanf("%d", &val);

            pos = search(e, n, val);

            if(pos == -1)
                printf("Employee not found\n");
            else
                printf("Employee found\n");

            break;

        case 4:
            scanf("%d", &val);
            n = deleteEmployee(e, n, val);
            break;

        case 5:
            return 0;
```

```
            default:
                printf("Invalid choice\n");
        }
    }
}
```

Output:

```
2
101 ram it 50000
102 joe finance 50000

1. Add Employee
2. Display Employees
3. Search Employee
4. Delete Employee
5. Exit
1
103 alice revenue 30000

1. Add Employee
2. Display Employees
3. Search Employee
4. Delete Employee
5. Exit
3
102
Employee found

1. Add Employee
2. Display Employees
3. Search Employee
4. Delete Employee
5. Exit
4
101

1. Add Employee
2. Display Employees
3. Search Employee
4. Delete Employee
5. Exit
2
102 joe finance 50000.00
```

103 alice revenue 30000.00

1. Add Employee
2. Display Employees
3. Search Employee
4. Delete Employee
5. Exit
5

10)#include<stdio.h>

```
float areoftri(int b, int h){
    return 1.0/2*b*h;
}

float areaofrect(int l, int b){
    return l*b;
}

int peroftri(int a, int b, int c){
    return a+b+c;
}

int perofrect(int l, int b){
    return 2*(l+b);
}

int main(){
    int l, b, ba, h, a, c;

    float (*f1)(int, int);
    float (*f2)(int, int);
    int (*f3)(int, int, int);
    int (*f4)(int, int);

    f1 = areoftri;
    f2 = areaofrect;
    f3 = peroftri;
    f4 = perofrect;

    printf("Enter length and breadth of rectangle: ");
    scanf("%d %d", &l, &b);
```

```
    printf("Area of rectangle = %.2f\n", f2(l, b));
    printf("Perimeter of rectangle = %d\n", f4(l, b));

    printf("Enter base and height of triangle: ");
    scanf("%d %d", &ba, &h);

    printf("Area of triangle = %.2f\n", f1(ba, h));

    printf("Enter three sides of triangle: ");
    scanf("%d %d %d", &a, &b, &c);

    printf("Perimeter of triangle = %d\n", f3(a, b, c));

    return 0;
}
```

Output:

```
Enter length and breadth of rectangle: 2 4
Area of rectangle = 8.00
Perimeter of rectangle = 12
Enter base and height of triangle: 4 5
Area of triangle = 10.00
Enter three sides of triangle: 3 4 5
Perimeter of triangle = 12
```